The Asymmetric Laplace Process: A Powerful Alternative to AL in Spatial Quantile Regression

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Abstract:

Quantile regression is a powerful tool for modelling the conditional distribution of a response variable, particularly when interest lies in the tails of the distribution. Standard quantile regression models typically employ the asymmetric Laplace (AL) distribution under the assumption of independent errors, and therefore cannot capture spatial dependence when it exists in the data. To address this, Lum and Gelfand (2012) introduced the Asymmetric Laplace Process (ALP), which incorporates a latent Gaussian process to capture spatial correlation within the error structure. We conduct a simulation-based comparison evaluating the ALP against the standard AL model under varying levels of spatial correlation. Results show that the ALP achieves lower RMSE for the regression parameter in almost all scenarios. However, credible interval coverage for β_0 declines substantially as the spatial range increases, ultimately falling below the nominal level. This deterioration in coverage suggests a potential identifiability issue between the spatial process and the intercept under strong spatial correlation. We conclude that while ALP offers improved point estimation, its credible intervals should be interpreted with caution in the presence of strong spatial dependence.

Keywords: Spatial quantile regression, Asymmetric Laplace process, Bayesian inference, Stan.

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1 Introduction

Quantile regression, introduced by Koenker and Bassett (1978), has become a fundamental tool for analysing the relationship between a response and a set of predictor variables across the entire conditional distribution. Unlike mean regression, which focuses solely on the conditional expectation, quantile regression allows researchers to investigate effects at different points of the response distribution, such as lower or upper tails. This flexibility has made quantile regression particularly valuable in fields such as economics, environmental science, public health, and real estate analysis, where understanding extreme outcomes is often of primary interest Koenker and Hallock (2001); Yu et al. (2003). A comprehensive treatment of the methodology is given in Koenker (2005).

A standard approach to Bayesian quantile regression employs the asymmetric Laplace (AL) distribution as the error term (Yu and Moyeed (2001)). The AL distribution is particularly attractive because maximizing its likelihood is equivalent to minimizing the check loss function, and its location parameter corresponds directly to the desired conditional quantile. Kozumi and Kobayashi (2011) further advanced this framework by introducing a convenient Gibbs sampling algorithm based on a location-scale mixture representation of the AL distribution, making Bayesian quantile regression computationally efficient and widely accessible.

Despite these advances, standard AL-based quantile regression models share a critical limitation: they assume independent errors. This assumption is often violated in many real-world applications where observations are collected across geographic locations. In spatial data, nearby observations tend to be more similar than distant ones, a phenomenon known as spatial autocorrelation. Ignoring such dependence can lead to biased parameter estimates, underestimated standard errors, and misleading inference (Banerjee et al. (2004)). To address this limitation, Lum and Gelfand (2012) proposed the Asymmetric Laplace Process (ALP). The ALP extends the standard AL model by embedding a latent Gaussian process within the error structure, thereby directly capturing spatial dependence. This approach preserves the interpretability of the regression coefficients as global quantile effects while allowing the errors to exhibit spatial correlation. The ALP has been shown to outperform non-spatial alternatives in applications such as housing price prediction and birth weight analysis (Lum and Gelfand (2012)).

However, the performance of the ALP under varying degrees of spatial correlation has not been systematically investigated. In particular, it remains unclear whether the advantages of the ALP hold uniformly across different spatial ranges and quantile levels. Moreover, potential identifiability issues between the spatial process component and the intercept parameter—especially under strong spatial dependence—have not been thoroughly examined. Understanding these limitations is crucial for practitioners who wish to apply the ALP to real spatially correlated data.

In this paper, we conduct a simulation-based comparison to evaluate the performance of the ALP against the standard non-spatial AL model of Kozumi and Kobayashi (2011). We focus specifically on the estimation accuracy and credible interval coverage of the regression parameters across different quantile levels and varying degrees of spatial correlation. Our results reveal a critical trade-off. On one hand, the ALP consistently achieves a lower root mean squared error (RMSE) compared to the non-spatial AL model, confirming the advantage of incorporating spatial structure to improve point estimation accuracy. On the other hand, we find that the credible interval coverage of the intercept under ALP declines substantially as spatial correlation becomes stronger. This deterioration suggests a potential identifiability problem between the spatial process and the intercept parameter when spatial dependence is high.

The remainder of this paper is organized as follows. Section 2 reviews the methodology. Section 3 presents the simulation design and results. The "Discussion and Conclusion" section ends with recommendations for applied researchers and directions for future work.

2 Methodology

2.1 Bayesian Quantile Regression

Let y_i be a scalar response and $\mathbf{x}_i \in \mathbb{R}^k$ a vector of predictors, $i = 1, \ldots, N$. The linear quantile model is specified as

$$y_i = \mathbf{x}_i^\top \boldsymbol{\beta}_p + \epsilon_i,$$

where $\boldsymbol{\beta}_p \in \mathbb{R}^k$ and ϵ_i satisfies $\Pr(\epsilon_i \leq 0) = p$.

Following Yu and Moyeed (2001), we assign $\epsilon_i \sim \text{ALD}(0, \tau, p)$ with density

$$f_p(\epsilon; \tau) = \tau p(1-p) \exp\{-\tau\, \rho_p(\epsilon)\},$$

where $\rho_p(u) = u(p - \mathbf{1}_{u<0})$ is the check loss function and $\tau > 0$ is the scale parameter. The ALD admits a location-scale Gaussian mixture (Kozumi and Kobayashi (2011)): for $\xi_i \sim \text{Exponential}(\tau)$

and $z_i \sim N(0,1)$ independently,

$$\epsilon_i = \theta\,\xi_i + \sqrt{\frac{2\xi_i}{\tau\,p(1-p)}}\,z_i, \qquad \theta = \frac{1-2p}{p(1-p)}. \tag{2.1}$$

Conditional on ξ_i, the error is Gaussian:

$$\epsilon_i \mid \xi_i \;\sim\; N\!\left(\theta\,\xi_i,\; \frac{2\xi_i}{\tau\,p(1-p)}\right).$$

2.2 The Asymmetric Laplace Process (ALP)

When observations are geo-referenced, Lum and Gelfand (2012) replace the independent scalar $z_i \sim N(0,1)$ in (2.1) with a mean-zero, unit-variance Gaussian process at location $\mathbf{s}_i$. Then the spatial error process becomes

$$\epsilon_p(\mathbf{s}) = \theta\,\xi(\mathbf{s}) + \sqrt{\frac{2\xi(\mathbf{s})}{\tau\,p(1-p)}}\,Z(\mathbf{s}),$$

where $\xi(\mathbf{s}) \overset{\text{iid}}{\sim} \text{Exponential}(\tau)$ and $Z(\mathbf{s})$ is a GP with mean zero and correlation function $\rho(\mathbf{s},\mathbf{s}')$. Each site retains an $\text{ALD}(0,\tau,p)$ marginal. Following Lum and Gelfand (2012), we decompose $Z(\mathbf{s}) = \sqrt{1-\alpha}\;w(\mathbf{s}) + \sqrt{\alpha}\;\delta(\mathbf{s})$, where $w(\mathbf{s})$ is a spatially structured GP with exponential correlation. Here, $\delta(\mathbf{s}) \overset{\text{iid}}{\sim} N(0,1)$ is a white noise process, and $\alpha \in [0,1]$. This ensures $\text{Var}[Z(\mathbf{s})] = 1$ for all $\mathbf{s}$.

2.3 Prior Specification

Following Lum and Gelfand (2012), the prior distributions are: $\beta_0,\ \beta_1 \sim N(0,\ 10^2)$; $\alpha \sim \text{Uniform}(0,1)$; $\phi \sim \text{Uniform}(\phi_{\min},\ \phi_{\max})$; $\tau \sim \text{Gamma}(a_\tau = 1,\ b_\tau = 1/\sqrt{2/[p(1-p)] + a_p})$. The prior bounds on ϕ are data-driven: $\phi_{\min} = \frac{3}{2\,d_{\max}}$ and $\phi_{\max} = \frac{3}{0.05\,d_{\max}}$, where $d_{\max}$ is the maximum pairwise distance.

2.4 MCMC Implementation

Both models are fitted in Stan (Stan Development Team , 2023) using Hamiltonian Monte Carlo with the No-U-Turn Sampler (Hoffman and Gelman , 2014), with 4 chains of 2000 iterations each (1000 warm-up), yielding 4000 post-warmup draws per fit. Convergence was assessed via the $\hat{R}$ statistic; all parameters satisfied $\hat{R} < 1.05$ across scenarios. To avoid the funnel geometry induced when sampling ξ_i jointly with τ, we apply a non-centred reparameterisation $\xi_i^{\text{raw}} = \tau\xi_i \sim \text{Exponential}(1)$ (Betancourt (2017)).

3 Simulation Design and Results

3.1 Data Generation

We generate $N = 50$ locations uniformly at random on $[0, 1] \times [0, 1]$ using the geoR package (Ribeiro and Diggle (2001)). This sample size is representative of moderate spatial datasets commonly encountered in practice and is consistent with the design of Lum and Gelfand (2012). For each of $S = 100$ replications, we simulate data across three quantile levels ($p \in \{0.10, 0.25, 0.50\}$) and three spatial ranges ($\phi \in \{0.20, 0.60, 0.90\}$), resulting in nine scenarios. The true parameters are set to $\beta_0 = -1.0$, $\beta_1 = 2.0$, and $\alpha = 0.20$.

The scale parameter τ for each quantile level is determined by requiring the marginal error variance to satisfy $\text{Var}(\epsilon_p) = 3.555$. Since $\text{Var}(\epsilon_p) = \left[\theta^2 + 2/\tau\, p(1-p)\right]/\tau$, where $\theta = (1 - 2p)/[p(1-p)]$, solving for τ yields $\tau_{0.10} \approx 0.490$, $\tau_{0.25} \approx 0.813$, and $\tau_{0.50} \approx 1.125$. This ensures a constant signal-to-noise ratio across quantile levels for fair comparison.

For each replication, we draw $\xi_i \overset{\text{iid}}{\sim} \text{Exponential}(\tau)$, sample a Gaussian random field $Z(\mathbf{s})$ with unit marginal variance and covariance

$$\text{Cov}[Z(\mathbf{s}_i), Z(\mathbf{s}_j)] = (1 - \alpha) \exp(-\|\mathbf{s}_i - \mathbf{s}_j\|/\phi) + \alpha\, \mathbf{1}_{i=j},$$

and set $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$, where $x_i \overset{\text{iid}}{\sim} N(0, 1)$ and

$$\epsilon_i = \theta\, \xi_i + \sqrt{\frac{2\xi_i}{\tau\, p(1-p)}}\, Z(\mathbf{s}_i).$$

3.2 Evaluation Metrics

We compare models using RMSE of the posterior mean, and empirical 95% coverage (nominal target: 0.95). We also use the mean in-sample check loss $\text{CL} = N^{-1} \sum_i \rho_p(\hat{u}_i)$ (Koenker and Bassett , 1978) and the Winkler interval score (Winkler , 1972) applied to the credible interval of β_0. For a nominal 95% credible interval (l, u) and true value y, the Winkler score at level $\alpha_W = 0.05$ is

$$W(l, u, y;\, \alpha_W) = (u - l) + \frac{2}{\alpha_W}(l - y)\, \mathbf{1}_{y<l} + \frac{2}{\alpha_W}(y - u)\, \mathbf{1}_{y>u}. \tag{3.1}$$

Lower scores are better. A narrow interval that covers the truth incurs only width as a penalty; an interval that misses the truth incurs an additional penalty of $2/\alpha_W = 40$ per unit of miss.

3.3 Results

Table 1 reports the performance metrics across all nine simulation scenarios for $p \in \{0.10, 0.25, 0.50\}$ and $\phi \in \{0.20, 0.60, 0.90\}$. For the RMSE and Winkler metrics, a ratio below 1.000 indicates that the spatial ALP model outperforms the AL model.

As observed in Table 1, the ALP model achieves lower RMSE than the AL model in every single scenario, for both the intercept (β_0) and the slope (β_1). The RMSE ratio decreases as p approaches 0.50 and as ϕ increases, reaching its minimum when the quantile level is $p = 0.50$ and the spatial range is moderate to strong ($\phi = 0.60$). This emphasizes that near the median of the response distribution and under strong spatial dependence, ignoring the spatial structure leads to loss of important information and increased estimation error. The slope estimator $\hat{\beta}_1$ follows the same ordering, although its ratios are more attenuated.

The most important finding concerns credible interval coverage. Both models show undercoverage for β_0 when the spatial range increases, but the degree of this problem is very different between them. The AL model has much lower coverage, falling to 0.25 at $(p, \phi) = (0.50, 0.90)$, because it assumes the residuals are independent while in reality they are spatially correlated. Because of this wrong assumption, the posterior underestimates the uncertainty of the parameters and the credible intervals become too narrow. The ALP coverage also deteriorates substantially, just less severely than AL — for example, it reaches 0.75 at $(p, \phi) = (0.50, 0.90)$, which is still well below the nominal level of 0.95. This remaining undercoverage in the ALP may be related to a potential identifiability problem between the global intercept and the latent spatial process, and this problem appears to become worse when the spatial dependence is stronger.

The Winkler interval score (3.1), a proper scoring rule that jointly penalises interval width and non-coverage, confirms this advantage: the ALP-to-AL ratio reaches 0.159 at $(p, \phi) = (0.50, 0.60)$, indicating that ALP-based credible intervals are simultaneously better calibrated and more informative. Conversely, the in-sample check loss ratio marginally exceeds unity across all scenarios (1.001–1.088), indicating that the ALP incurs a slight additional in-sample error attributable to spatial regularisation; however, as reported by Lum and Gelfand (2012), the ALP demonstrates greater advantages in out-of-sample prediction.

The bias was approximately zero for both models across all scenarios, although the AL model performed slightly better on this metric; these results are omitted from Table 1 for the sake of brevity. As shown in Table 1, the credible interval coverage for β_1 ranged between 0.91 and 0.98 across all scenarios, with the ALP showing a slight overall advantage.

Table 1: Simulation performance metrics comparing the Asymmetric Laplace Process (ALP) and the non-spatial Asymmetric Laplace model (AL) across nine scenarios defined by quantile level p and spatial range ϕ. RMSE and Winkler ratios are computed as ALP/AL. An RMSE or Winkler ratio below 1.000 indicates superior performance for the ALP model. Empirical 95 % credible interval coverage is reported for β_0 and β_1; the nominal level is 0.95. CL = check loss.

		RMSE Ratio (ALP/AL)		95 % Coverage (β_0)		95 % Coverage (β_1)		Winkler (β_0)	CL
p	ϕ	$\hat{\beta}_0$	$\hat{\beta}_1$	ALP	AL	ALP	AL	Ratio	Ratio
	0.2	0.959	0.990	**0.90**	0.83	**0.98**	0.97	0.894	1.002
0.10	0.6	0.930	0.999	**0.71**	0.56	0.95	0.95	0.637	1.001
	0.9	0.923	1.006	**0.62**	0.44	0.95	**0.96**	0.591	1.002
	0.2	0.857	0.864	**0.86**	0.57	0.95	0.95	0.476	1.008
0.25	0.6	0.766	0.828	**0.71**	0.31	**0.96**	0.95	0.318	1.038
	0.9	0.784	0.842	**0.64**	0.29	**0.96**	0.93	0.352	1.040
	0.2	0.699	0.804	**0.86**	0.55	0.94	**0.95**	0.327	1.023
0.50	0.6	0.594	0.702	**0.82**	0.26	**0.96**	0.91	0.159	1.076
	0.9	0.635	0.697	**0.75**	0.25	**0.95**	0.91	0.211	1.088

Discussion and Conclusion

While accounting for spatial dependence via the ALP Lum and Gelfand (2012) substantially improves point estimation accuracy over the non-spatial AL model Kozumi and Kobayashi (2011), our findings reveal a critical trade-off in uncertainty quantification. Specifically, as the spatial range increases, the severe drop in credible interval coverage for the intercept suggests a potential identifiability issue between the global intercept and the latent spatial process. Therefore, while the ALP is highly effective for point estimation, applied researchers must interpret its credible intervals with caution under strong spatial correlation.

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